

ScentCarving: Fabricating Thin, Multi-layered and Paper-Based Scent Release through Laser Printing

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Figure 1: (A) The thin and multi-layered structure of *ScentCarving*. (B) Engraving the heat-responsive embossing powder onto the paper using a CNC machine. (C) Intriguing, colorful, and Scent-enhanced tableware devised for guiding children's eating behavior. (D) Recreation of Van Gogh's "Sunflowers" opening the petal while releasing the fragrances. (E) Design explorations for scent release (smell intensity and odor mixing) and graphics.

ABSTRACT

Differentiating from commonly-seen Olfactory Displays (ODs) that utilize rigid mechanisms within HCI, we propose *ScentCarving*, a controllable, lightweight, and flexible odor-release mechanism through a thin, multi-layered structure. *ScentCarving* consists of four layers, including the (1) *paper-based substrate*, (2) *odor layer* containing the ink-based aroma, (3) *odor-sealing layer* by introducing a thermoplastic material, and (4) a *heating module* using conductive cooper to soften the sealing layer and release the scent. *ScentCarving* also involves an easy-to-access Odor Printing technique capable of engraving fragrances onto thin and flexible substrates. We conducted a small-scale user study to test the scent-releasing behavior in terms of *smell distance* and *responsive time*.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**.

KEYWORDS

smell printing; paper-based olfactory display; shape-changing; thermoplastic; odor ink; fabrication; Human-smell Interaction

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1 INTRODUCTION

In Human-Computer Interaction (HCI), the sense of smell features in arousing memories [2], affecting emotions [3] and cognition [8], maintaining appetite [16, 17], attention [7], and mental health [14], and enhancing social interactions [18], educational experiences [9], and learning [15]. However, due to the intrinsic properties of odor (e.g., requiring odor sources), existing odor-release mechanisms were often limited within rigid and cumbersome designs [1, 6, 8, 10, 13, 22]. To address this challenge, we aim to increase the flexibility and aesthetics of scent-releasing methods. Besides, as an easy-accessible, lightweight, and flexible material, paper is commonly utilized in various fields, such as printing, packaging, art, and education. HCI researchers and designers also favored this material and created many paper-based interactive objects [12, 19–21]. Previous research has attempted to incorporate smells into thin sheet materials, for example, creating low-fidelity olfactory experiences through scratching [4], utilizing inkjet printers to produce graphics with natural scent emission [11] within Computer Graphics (CG), and proposing geranium-inspired encapsulation arrays for controlled odor release within a professional chemical laboratory [5]. Inspired by these works, we propose *ScentCarving*, a controllable, lightweight, and flexible odor-release mechanism

through a thin, multi-layered structure and an accessible fabrication pipeline.

2 SCENTCARVING

Thin and multi-layered Scent-Releasing Mechanism. As shown in Figure 1A, the structure mainly consists of four layers. (1) For the *substrate layer*, we used paper-based materials with various weights, such as thin sulfite paper (Figures 1B and D) and copy paper (60g), kraft paper (120g), and 0.1mm thick eco-friendly paper sauce (Figure 1C). (2) The *odor layer* contains and reserves the aromas. We chose less odorous screen printing ink (Merlia, WPET-102) as the main solvent to dilute the scented source for its excellent adhesiveness when combining with paper-based materials. We recommend using water-prove substrates for better adhesive performance. (3) For the *odor-sealing layer*, we introduce a thermoplastic material, Polyamide resin powder (embossing powder), on top of the odor layer to prevent the scent from leaking. This material states solid without heating for odor sealing while alternating to soft (with expanded molecule space) when exposed to thermal activation (130°C). This can benefit the acceleration of delivering the scented molecules. (4) We used 2mm wide copper tape (0.12 Ω/cm) as the *heating circuit layer* attached to the backside of the substrate (Figure 1C). The joule heat increases the temperature from 27°C to 70°C in 8.5 seconds when 1 Voltage power and 5.2A current are applied. During the fundamental material test, we found that the thin structure gets curved (from 50° to 160°) when using the substrates with a thickness lower than 150g (Figure 1D).

Accessible Fabrication Pipeline. *Step 1:* Firstly, we mixed the water-soluble essential oils with transparent ink in a 1:1 ratio. Then, we manually transformed the scented ink onto the substrate using screen printing with a predefined graphic pattern. Before the ink gets cured, we need to cover and stick the embossing powder evenly onto the odor layer. *Step 2:* We use the heat generated by the laser engraving machine to melt the embossing powder by setting the cutting speed to 20mm/s with 250 mW, which will shape it into a thin film covering the entire odor layer. In this step, we also tried a heat gun to melt the powder; however, the high-temperature wind released the scent with a strong concentration, while the laser cutter benefited from its high speed and micro heating area. *Step 3:* Finally, we affixed the conductive copper strips to the backside of the substrate based on the pre-devised graphics and connected it to the power source to generate the controllable heat.

3 SCENT RELEASING TEST

We tested the *smell distance* and *responsive time* in a workspace with open windows at 25°C room temperature and humidity of 62%rh. Eight participants (4 females) with an average age of 24 (SD = 1.31) signed the consent form with no one was allergic and sensitive to odors. For the test setup, we first prepared five square and orange-scented samples with varying areas listed in Table 1. We placed the swatches horizontally on five separate tables and connected them to the power with a 5A current for heating 8s. The participants were required to walk to the samples sequentially and stop when they detected the odor (slowly getting closer to the sample until they had an apparent smell perception). The experimenters measured the distance with a ruler and recorded the data. Secondly, we

tested the responsive time of the five samples by setting a certain distance (approximately 20cm) between the users and samples. The participants were required to report the time when they smelled the scents.

Results and Discussion. Table 1 shows our results. We found that even with a small area of the scented source, we could still achieve a relatively long distance (25.8cm). Therefore, we recommend using less than 4cm² of scented graphics to create close-to-nose smell-enhanced designs. We also found that *ScentCarving* could potentially create environmental and ambient smell experiences by devising an aromatic pattern larger than 100cm². Besides, even though *ScentCarving* needs a relatively long smell responsive time (from 9.4s to 14.5s from the test), we could also utilize and integrate with other features that *Scentcarving* afforded, such as the bendability and foldability, to create dynamic scent release through the soft mechanical design (e.g., origami and kirigami). Additionally, we recommend avoiding direct skin contact with the heating circuits for safety considerations, and the structure can be optimized by adding thermo-insulated materials.

Table 1: As the graph area increases, the response time decreases, and the maximum smell distance increases.

Samples	Response time	Maximum Smell Distance
A: 4 cm ²	14.5s	25.8 cm
B: 16 cm ²	12.1s	31.2 cm
C: 36 cm ²	10.5s	43.2 cm
D: 64 cm ²	10.2s	53.8 cm
E: 100 cm ²	9.4s	81.4 cm

4 APPLICATIONS AND FUTURE WORK

We envision that *ScentCarving* can seamlessly and aesthetically merge into our daily settings by devising complex, delicate, and dynamic odor graphics. As shown in Figure 1C, we could create scented tableware to manipulate people's appetites with decreased taste (e.g., children and elders). We could also add a smell-enhanced feature for Shape-Changing Interfaces (SCIs) to create dynamic and changeable smell experiences by utilizing the deformation properties of thermoplastic materials, such as artistic sunflowers for notification (Figure 1-D), interactive educational books (e.g., flora and plants), paper-based rehabilitation equipment, and scented origami toys. For future work, we will keep improving *ScentCarving* through material iteration and design explorations. For instance, we will replace Polyamide resin by selecting user-friendly and food-grade materials (e.g., gelatin powder and sugar), and conduct rigorous smell experience experiments with more complex structures (e.g., double-sided and overlay structures).

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REFERENCES

- [1] Judith Amores, Javier Hernandez, Artem Dementyev, Xiqing Wang, and Pattie Maes. 2018. Bioessence: A wearable olfactory display that monitors cardio-respiratory information to support mental wellbeing. In *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 5131–5134.
- [2] Judith Amores Fernandez, Nirmita Mehra, Bjoern Rasch, and Pattie Maes. 2023. Olfactory Wearables for Mobile Targeted Memory Reactivation. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–20.
- [3] Giada Brianza, Jesse Benjamin, Patricia Cornelio, Emanuela Maggioni, and Marianna Obrist. 2022. QuintEssence: a probe study to explore the power of smell on emotions, memories, and body image in daily life. *ACM Transactions on Computer-Human Interaction* 29, 6 (2022), 1–33.
- [4] Jas Brooks and Pedro Lopes. 2023. Smell & Paste: Low-Fidelity Prototyping for Olfactory Experiences. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [5] Bingda Chen, Meng Su, Qi Pan, Zeyang Zhang, Shuoran Chen, Zhandong Huang, Zheren Cai, Zheng Li, Xin Qian, Xiaotian Hu, et al. 2019. Fully Printed Geranium-Inspired Encapsulated Arrays for Quantitative Odor Releasing. *ACS omega* 4, 22 (2019), 19977–19982.
- [6] Yongsoon Choi, Adrian David Cheok, Xavier Roman, The Anh Nguyen, Kenichi Sugimoto, and Veronica Halupka. 2011. Sound perfume: designing a wearable sound and fragrance media for face-to-face interpersonal interaction. In *Proceedings of the 8th international conference on advances in computer entertainment technology*. 1–8.
- [7] Dmitrijs Dmitrenko, Emanuela Maggioni, Giada Brianza, Brittany E Holthausen, Bruce N Walker, and Marianna Obrist. 2020. Caroma therapy: pleasant scents promote safer driving, better mood, and improved well-being in angry drivers. In *Proceedings of the 2020 chi conference on human factors in computing systems*. 1–13.
- [8] David Dobbstein, Steffen Herrdum, and Enrico Rukzio. 2017. inScent: A wearable olfactory display as an amplification for mobile notifications. In *Proceedings of the 2017 ACM International Symposium on Wearable Computers*. 130–137.
- [9] M Garcia-Ruiz, Samir A El-Seoud, Arthur Edwards, Jihad Mohamad AL-JA'AM, and Raul Aquino-Santos. 2008. Integrating the sense of smell in an educational human-computer interface. *Interactive Computer Aided Learning* (2008).
- [10] You-Yang Hu, Yao-Fu Jan, Kuan-Wei Tseng, You-Shin Tsai, Hung-Ming Sung, Jin-Yao Lin, and Yi-Ping Hung. 2021. abio: Active bi-olfactory display using subwoofers for virtual reality. In *Proceedings of the 29th ACM International Conference on Multimedia*. 2065–2073.
- [11] Yi-Ching Kang and Hiroki Nishino. 2019. Scented graphics: exploration in inkjet scented-printing. In *ACM SIGGRAPH 2019 Posters*. 1–2.
- [12] Naoya Koizumi, Kentaro Yasu, Angela Liu, Maki Sugimoto, and Masahiko Inami. 2010. Animated paper: A toolkit for building moving toys. *Computers in Entertainment (CIE)* 8, 2 (2010), 1–16.
- [13] Yuxuan Lei, Qi Lu, and Yingqing Xu. 2022. O&O: A DIY toolkit for designing and rapid prototyping olfactory interfaces. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–21.
- [14] Zilan Lin, Kai Kunze, Atsuro Ueki, and Masa Inakage. 2020. Aromacue-a scent toolkit to cope with stress using the 4-7-8 breathing method. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction*. 265–272.
- [15] Qi Lu, Yuewei Zhang, Yuxin Zhang, Shao-En Ma, Yunfan Zhang, Yuou Qin, Peizhong Gao, Qiwei Zhang, and Yingqing Xu. 2023. Atmospheror: Towards an Olfactory Interactive System for Enhancing Social Presence and Interaction in Synchronous Online Classes. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–8.
- [16] Daiki Mayumi, Yugo Nakamura, Yuki Matsuda, Shinya Misaki, and Keiichi Yasumoto. 2022. Aromug: Mug-type olfactory interface to assist in reducing sugar intake. In *Adjunct Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing and the 2022 ACM International Symposium on Wearable Computers*. 183–187.
- [17] Daiki Mayumi, Yugo Nakamura, Yuki Matsuda, Kentaro Ueda, Shinya Misaki, and Keiichi Yasumoto. 2023. Kaolid: Lid-type Olfactory Interface to Improve Taste of Beverages with Ortho-Retronasal Smell. In *2023 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*. IEEE, 303–305.
- [18] Siddharth Mehrotra, Anke Brouck, Marianna Obrist, and Jan Borchers. 2022. The Scent of Collaboration: Exploring the Effect of Smell on Social Interactions. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–7.
- [19] Masa Ogata and Masaaki Fukumoto. 2015. Fluxpaper: Reinventing paper with dynamic actuation powered by magnetic flux. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 29–38.
- [20] Simon Olberding, Sergio Soto Ortega, Klaus Hildebrandt, and Jürgen Steimle. 2015. Foldio: Digital fabrication of interactive and shape-changing objects with foldable printed electronics. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology*. 223–232.
- [21] Guanyun Wang, Tingyu Cheng, Youngwook Do, Humphrey Yang, Ye Tao, Jianzhe Gu, Byoungkwon An, and Lining Yao. 2018. Printed paper actuator: A low-cost reversible actuation and sensing method for shape changing interfaces. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [22] Yanan Wang, Zhitong Cui, Hebo Gong, and Ting Chen. 2023. OlfacKit: A Toolkit for Integrating Atomization-Based Olfactory Interfaces into Daily Scenarios. *International Journal of Human-Computer Interaction* (2023), 1–20.